Impact of Weather Conditions on Flight Delays in the United States

Data Visualisation and Exploration Project Report (DVP Part 2)

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1. Introduction

Adverse weather remains one of the most significant sources of flight disruption in global aviation. This project, *Impact of Weather Conditions on Flight Delays in the United States*, investigates how three key meteorological variables—rain, wind, and snow—affect the percentage of delayed flights across three major U.S. hubs: **Hartsfield–Jackson Atlanta International Airport (ATL)**, **Newark Liberty International Airport (EWR)**, and **John F. Kennedy International Airport (JFK)**. The analysis spans 2019–2024 and integrates operational flight-performance records from the **Bureau of Transportation Statistics (BTS)** with meteorological data from the **National Oceanic and Atmospheric Administration (NOAA)**, forming a robust multi-year dataset suitable for interactive exploration.

The visualisation seeks to communicate complex temporal and categorical relationships through an engaging, story-driven dashboard built with D3 (v7). Its purpose is to transform raw quantitative data into an intuitive narrative that reveals *how environmental variability shapes aviation punctuality*. Specifically, the project addresses three research questions:

- **RQ_1:** How do different weather types (rain, wind, snow) influence the frequency of flight delays?

- **RQ_2:** Which airports experience the highest weather-related delay percentages, and how do these patterns compare temporally and spatially?

- **RQ_3:** How do seasonal trends and yearly variations modulate delay intensity across airports?

The intended audience comprises aviation analysts, airport-operations planners, meteorological researchers, and policy specialists who rely on data-driven insights rather than static tabular summaries. Following **Munzner's (2014)** *what–why–how* analytical framework, the visualisation supports both *comparison* and *temporal-pattern analysis* through coordinated views—bar, line, and heat-map charts—that maintain visual consistency and reinforce cognitive fluency. Design choices were guided by perceptual principles from **Ware (2021)** and minimal-cognitive-load guidelines from **few (2012)**, ensuring clarity and rapid pattern recognition.

Each chart within the system functions not as a standalone graphic but as an inter-linked component of a coherent narrative (Segel & Heer, 2010). Linked interactions—hover highlighting, synchronized fading, quick-view presets, and dropdown filters—allow users to interrogate cause–effect relationships across weather variables and airports. This approach aligns with **Shneiderman's (1996)** mantra *"overview first, zoom and filter, then details on demand"*, ensuring progressive disclosure of information without overwhelming the viewer. Accessibility and usability considerations followed **Nielsen's (2012)** heuristics and **WCAG 2.1 (W3C, 2018)** standards through careful colour contrast, typographic hierarchy, and annotation placement.

In summary, this project transforms large-scale transportation and meteorological data into an interactive narrative experience that reveals how shifting atmospheric conditions propagate delays across the U.S. air-traffic network. The following section documents the **Five Design Sheets (FdS)** process (Roberts, 2011; Mazza, 2009), outlining the iterative evolution from conceptual sketches to the final, theoretically informed dashboard.

2. Design Process

2.1 Overview of the Five Design Sheets Approach

The design of the *Impact of Weather on U.S. Flight Delays* visualisation followed the **Five Design Sheets (FdS)** framework (Roberts et al., 2014), a structured methodology supporting creative exploration, iteration, and theory-driven refinement. Each sheet progressively evolved from open ideation to a coherent interactive narrative. The process was informed by **Munzner's (2014) What–Why–How model**, **Kirk's (2016)** design workflow, **Gestalt perceptual principles** (Ware, 2021), and **Tufte's (2001)** emphasis on clarity and minimalism. Together, these guided decisions on layout, interaction, and cognitive flow, ultimately producing a visually unified and analytically powerful dashboard.

2.2 Design Sheet 1: Concept Ideation

Sheet 1 captured the **brainstorming stage**, generating broad ideas on how to convey relationships between weather variables and flight delays.Initial sketches explored several directions:

- Comparative bar and line charts showing temporal delay trends.
- A **geospatial map** overlaying airports, delay intensity, and weather variables (using bubble or heatmap layers).
- Combinations of time-series, categorical, and spatial views.

While the geospatial and bubble-heatmap concepts offered potential for geographical storytelling, they were ultimately considered **less feasible** within project constraints and did not directly support analytical clarity. The brainstorming reflected Munzner's *"Why"* stage—defining audience goals (aviation analysts and policy planners) and analytical tasks (comparison, correlation, temporal analysis). Gestalt's **proximity and similarity** principles (Ware, 2021) informed early clustering of weather categories, laying the groundwork for later coordinated charts.

2.3 Design Sheet 2: Alternative Layout and Visual Encodings

Sheet 2 transitioned from free exploration to a **linear narrative layout**, presenting data in sequential panels—overall statistics, monthly analysis, and airport comparisons. A major new idea was the **inclusion of a heatmap** to visualise intensity of delays by airport and weather type. The design emphasised a top-to-bottom reading order, guiding viewers through the story much like scrolling articles described by Kirk (2016).

However, evaluations showed that linear scrollytelling limited user interactivity and made it difficult to cross-compare weather types simultaneously. Consequently, the heatmap idea was retained conceptually but planned for integration into a **dashboard format** in later sheets. This stage advanced the "How" layer of Munzner's framework—experimenting with encoding strategies and narrative sequence.

2.4 Figure 3 – Design Sheet 3: Narrative Flow and Interaction Plan

Sheet 3 introduced a **grid-based dashboard layout**, replacing the linear flow with an interactive, multi-panel environment. This version pioneered the **annotation-box concept**, designed to appear beneath each chart and dynamically update with hover actions—transforming quantitative graphics into contextual narratives. Quick-view **presets for Rain, Wind, and Snow** were also introduced to simplify comparisons.

These interactions embodied **Shneiderman's (1996)** "overview → filter → details on demand" principle, while **North's (2000)** coordinated-multiple-views idea informed the coupling of charts. Hover events dimmed unrelated visuals, creating perceptual focus via Gestalt's figure-ground separation. Though conceptually strong, some spacing and text-placement issues emerged when tested on laptop screens, motivating layout adjustments in the next iteration.

2.5 Figure 4 – Design Sheet 4: Drill Down Approach.

Sheet 4 evolved into a **drill-down style layout**, allowing users to explore deeper insights for specific airports or weather categories. It incorporated **dedicated tiles** describing *data sources* and *dataset attributes*—enhancing transparency and aligning with Kirk's (2016) emphasis on trust and metadata visibility.

This version adopted a cleaner aesthetic inspired by **Tufte's (2001)** data-ink ratio principle—retaining only essential elements while ensuring legibility. The design successfully balanced analytical richness with information hierarchy, though it required integration with the stronger annotation and preset concepts of Sheet 3. Thus, Sheet 4 served as a conceptual bridge between analytical depth and narrative guidance.

2.6 Figure 5 – Design Sheet 5: Final Design Specification

Sheet 5 synthesised the most effective components from previous sheets into a single **unified, interactive dashboard**:

- **From Sheet 2:** Overall layout and inclusion of the **heatmap** for categorical intensity.
- **From Sheet 3: Annotation boxes** and **quick-view presets** to enable dynamic storytelling.
- **From Sheet 4: Data-description and source tiles** ensuring transparency and contextual framing.

The result was a **tri-panel layout** featuring:

- **Left Panel:** Quick-view presets (Rain / Wind / Snow).
- **Centre Panel:** Three coordinated charts — *bar chart* (airport × weather), *line chart* (monthly trend 2019–2024), *heatmap* (aggregate delay severity).

- **Right Panel:** Filters for airports and weather, plus dataset information.

All charts were synchronised via **D3.js v7** data binding, ensuring real-time interactivity: selecting a weather type or airport simultaneously updated every chart. Smooth **three-second transitions** maintained perceptual continuity (Heer & Robertson, 2007). Annotation boxes changed dynamically on hover—e.g., "ATL recorded 24.8% delays under rain"—linking visual and textual narratives (Hullman & Diakopoulos, 2011).

The design conformed to Munzner's framework:

- **What:** Multivariate temporal data (delay %, weather type, and airport).
- **Why:** Identify and compare weather-related delay patterns.
- **How:** Linked D3 visual encodings with animation and coordinated filtering.

2.7 Summary of Design Evolution

The final implementation integrated core data-visualisation theories:

- **Consistency & Hierarchy:** Repeated hover logic, axis style, and typography ensured a predictable visual grammar (Few, 2012; Norman, 2013).
- **Colour & Perception:** Blue (Rain), Red (Wind), Purple (Snow) enabled preattentive grouping (Ware, 2021) while maintaining aesthetic harmony.
- **Interaction & Narrative:** Hover tooltips, fading context, and annotations embodied *interactive storytelling* best practices (Shneiderman, 1996; Hullman & Diakopoulos, 2011).
- **Visual Clarity:** Simplified panels, clean typography, and reduced ornamentation maximised data-ink ratio (Tufte, 2001).
- **Adaptability:** CSS responsiveness ensured readability across laptops and desktops without loss of context.

By combining the structured creativity of the Five Design Sheets with theoretical grounding, the final product achieved narrative coherence, analytical precision, and visual polish—transforming complex aviation–weather data into an intuitive, story-driven analytical experience.

3. Implementation

3.1 Technical Implementation

The *Impact of Weather on U.S. Flight Delays* narrative visualisation was fully implemented using **HTML5**, **CSS3**, and **JavaScript** with the **D3.js v7** library. This combination enabled highly interactive, browser-native data storytelling without requiring external frameworks. The project adopted a modular structure — index.html (layout and content), style.css (visual styling and responsiveness), and main.js (data processing and interactivity) — ensuring a clear separation of presentation and logic consistent with web-engineering best practices.

The dataset, *flights_weather_JFK_EWR_ATL_clean.csv*, integrated flight performance records from the **U.S. Bureau of Transportation Statistics (BTS)** and meteorological data from the **National Oceanic and Atmospheric Administration (NOAA)** between **2019 and 2024**. Each entry contained fields such as airport_code, date, arr_delay_15m, dep_delay_15m, and binary indicators for weather conditions (rain_flag, snow_flag, windy_flag). Using D3's d3.csv () and d3.rollups () functions, the raw data was aggregated to compute average delay percentages by airport, weather type, and month, producing a cleaned, analysis-ready dataset.

The implementation **faithfully executed the complete layout and functionality defined in Sheet 5**, realising its **tri-panel interactive design** and **narrative coordination** in full.

- **Left Panel – Quick Views:** Implemented with responsive buttons allowing users to instantly filter by weather category (Rain, Wind, or Snow).
- **Centre Panel – Visualisation Hub:** Integrated three coordinated charts:
    1. A **Bar Chart** comparing delay percentages across airports and weather conditions.
    2. A **Line Chart** showing monthly delay patterns (2019–2024).
    3. A **Heatmap** visualising aggregated delay intensity by airport–weather pairings.
- **Right Panel – Controls and Information:** Included dropdowns for airport and month selection, as well as a **data description and source box** summarising the BTS–NOAA integration and temporal coverage.

Each visual component was linked through a shared global data array (data []), enabling **synchronous updates** — so a single user interaction automatically propagated across all charts. This approach operationalised **North's (2000)** *Coordinated Multiple Views (CMV)* model, allowing users to cross-compare trends across multiple perspectives simultaneously.

Interactivity and narrative layering were implemented through D3 event listeners and transition functions. Hovering over any data point activated two complementary feedback mechanisms:

1. A **tooltip** adjacent to the cursor, presenting precise delay values and contextual information.
2. A **contextual annotation box next to each chart**, dynamically updating narrative text based on the hovered element.

These annotation boxes, derived from Sheet 3's conceptual model and integrated within Sheet 5's structure, strengthened narrative coherence by directly linking quantitative visuals with explanatory insight. The interface also applied **fading logic**—when a bar or node was hovered, surrounding elements dimmed—to maintain user focus while preserving contextual awareness.

Animated transitions (.transition ().duration (3000).ease (d3.easeCubicOut)) were incorporated to render visual elements gradually within three seconds. This animation approach, following **Heer and Robertson (2007)**, enhanced perceptual continuity and prevented abrupt context shifts. The resulting interaction created a seamless and engaging analytical journey where users could explore how different weather phenomena correlated with flight delays across airports and months.

The styling followed **Tufte's (2001)** minimalist philosophy and **Few's (2012)** principles of data clarity, using a semi-transparent white overlay and soft shadows to distinguish sections without visual clutter. **CSS media queries** ensured the dashboard automatically scaled for both laptop and desktop screens while maintaining consistent proportions.

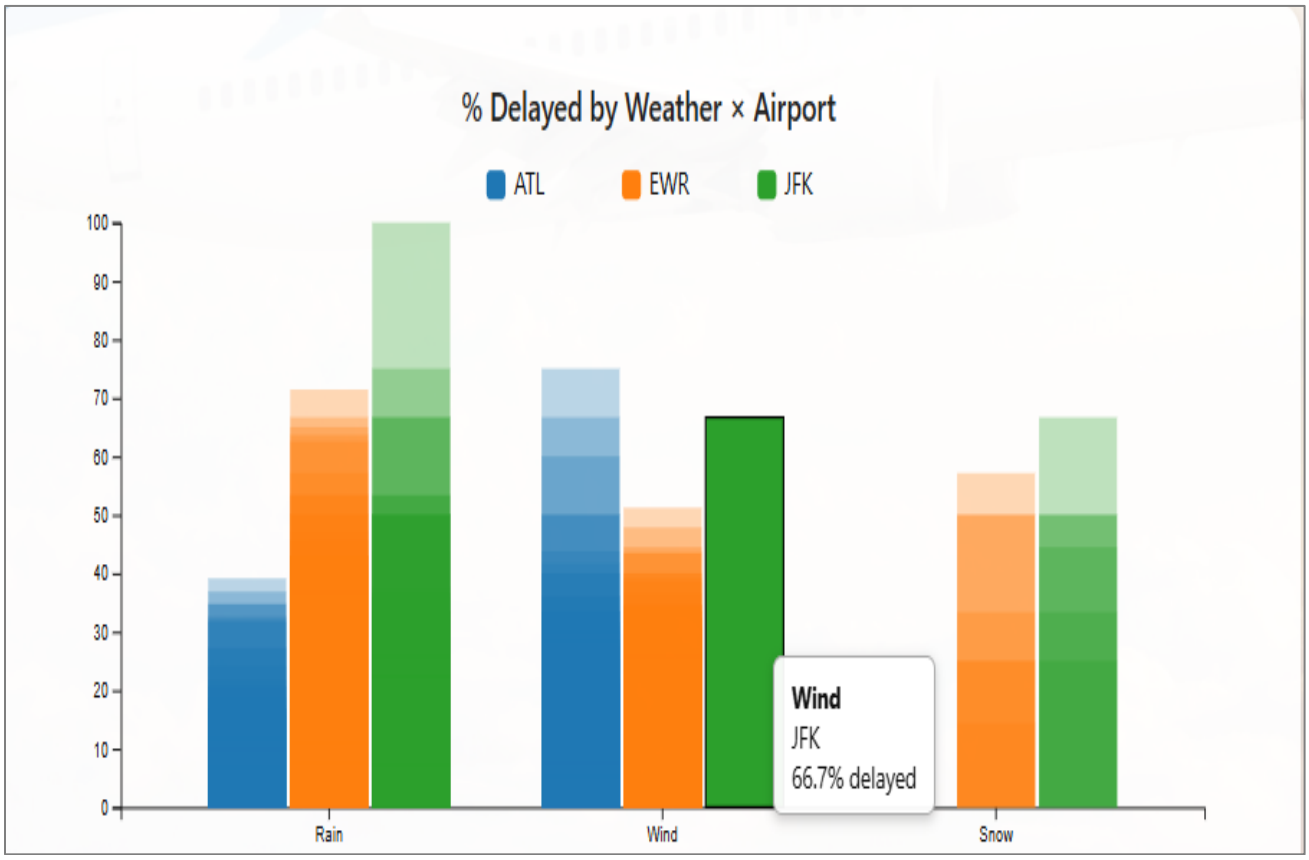
Minor implementation adjustments were introduced only for **performance optimisation and device compatibility**. For example, the earlier geospatial bubble concepts were omitted after testing due to limited interpretive benefit and increased rendering load. Tooltip placement and spacing were fine-tuned to maintain alignment under varying resolutions.

Ultimately, this implementation **realised the complete design intent of Sheet 5**, uniting analytical depth with visual storytelling. The coordinated animations, contextual annotations, and dynamic synchronisation achieved an interactive narrative system that effectively communicates the relationship between meteorological conditions and flight delays across major U.S. airports.

3.2 Interactive Narrative Visualisation Implementation

The implemented narrative visualisation consists of three interlinked components — a **Bar Chart**, a **Line Chart**, and a **Heatmap** — each contributing a distinct analytical layer within a unified storytelling framework. The bar chart provides a categorical comparison of weather-induced delays across major U.S. airports, establishing an immediate visual overview. The line chart introduces a temporal dimension, illustrating monthly delay trends from 2019 to 2024 and enabling the exploration of seasonal weather impacts. The heatmap complements these by revealing the intensity and co-occurrence of weather effects across airports and conditions. All three visualisations are **synchronised in real time**: selections or hovers on any chart dynamically update the others, supported by contextual annotation boxes and coordinated transitions that maintain narrative flow and analytical coherence.

3.2.1 Bar Chart – Weather-Based Delay Comparison



(Figure 3.2.1 — Bar Chart: Weather-Based Delay Comparison)

The **bar chart** in **Fig 3.2.1** serves as the primary visual gateway of the narrative, establishing a clear comparison of flight delay percentages across weather types (*Rain*, *Wind*, *Snow*) and major airports — **Hartsfield–Jackson Atlanta International (ATL)**, **Newark Liberty International (EWR)**, and **John F. Kennedy International (JFK)**. This chart functions as both an overview and an entry point for exploration, inviting users to examine which airports are most vulnerable under specific weather conditions.

Purpose and Narrative Role

The bar chart introduces the viewer to the project's central question — *how do different weather types affect flight delays across major U.S. airports?* It supports rapid pattern recognition by presenting grouped comparisons in a single frame, enabling users to identify dominant delay factors without scrolling or toggling between charts. As the first visual in the dashboard, it provides a high-level anchor from which users can drill down into temporal or spatial trends in subsequent charts.

Data Encoding and Colour Logic

Delay percentages are encoded on the **y-axis**, while **weather categories (Rain, Wind, and Snow)** occupy the **x-axis**. Each group contains three bars, representing **ATL (blue)**, **EWR (orange)**, and **JFK (green)**. This airport-based colour scheme was selected for categorical clarity and accessibility, ensuring that the same hue consistently represents each airport across all visualisations. The palette adheres to perceptual contrast guidelines to maintain readability against the semi-transparent white dashboard panels.

A **horizontal baseline** aids magnitude comparison, and a **legend positioned below the title** clarifies the airport–colour association without consuming excess space. The use of moderate saturation levels reduces visual fatigue while maintaining distinctiveness across categories.

Interactive Behaviour

Hovering over any bar activates a **tooltip overlay** that provides contextual detail — airport name, weather condition, and exact percentage delayed. Simultaneously, the selected bar highlights at 100 % opacity, while the others fade to ~40 %, reinforcing the principle of **focus + context**. The animation timing is tuned for smoothness and responsiveness across devices, ensuring the user can explore without perceptible lag.

The interaction is **linked** with the other visualisations: selecting a weather type dynamically updates the line chart and heatmap to display temporal and spatial trends for that condition. This coordination creates a cohesive narrative flow, allowing viewers to trace patterns from summary (bar chart) to evolution (line chart) to distribution (heatmap).

Design and Layout Considerations

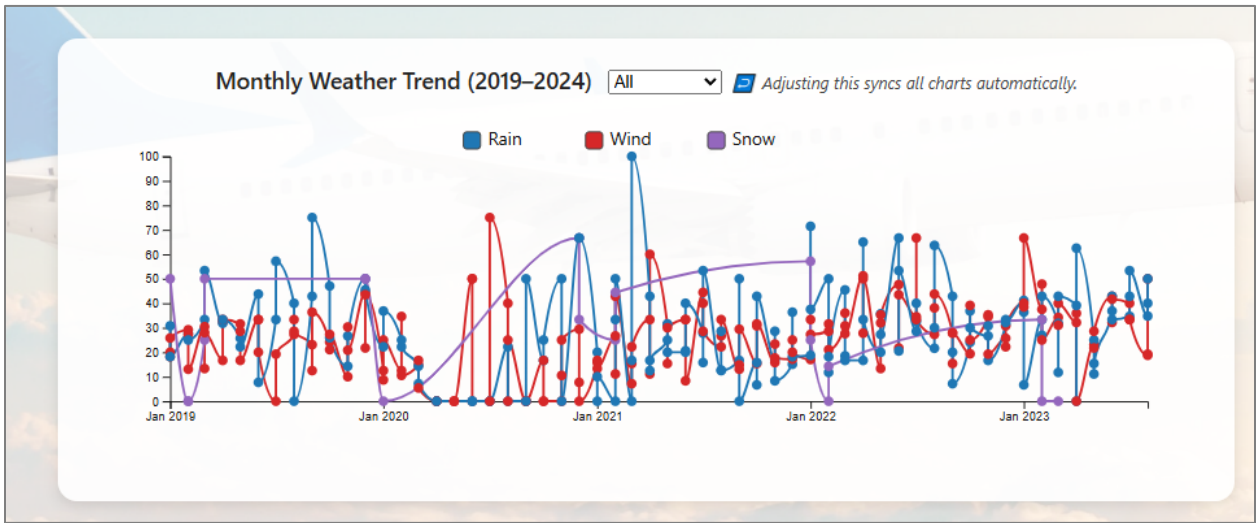
The chart title — *"% Delayed by Weather × Airport"* — employs concise wording and direct variable notation to strengthen readability. Axis labels are intentionally minimal, focusing attention on the proportional differences rather than exact numeric values. The layout maintains generous white space around the grouped bars to prevent crowding and allow perceptual grouping by weather category.

The choice of a grouped-bar layout follows Cleveland and McGill's (1984) findings that position along a common scale yields higher accuracy in comparing magnitudes than alternatives like stacked bars or pies. Combined with subtle hover animation and real-time legend feedback, the design balances visual simplicity with analytical depth.

Summary

Overall, the bar chart provides the dashboard's **comparative foundation** — a static entry view that becomes interactive when the user engages with it. It clearly communicates which airports experience the greatest percentage of delays under each weather type and prepares the viewer to investigate *when* and *why* those variations occur in the following visual components.

3.2.2 Line Chart – Monthly Weather Trends (2019–2024)



(Figure 3.2.2 — Line Chart: Monthly Weather Trends 2019–2024.)

The **line chart** in Fig3.2.2 extends the comparative insight of the grouped bar chart by introducing a **temporal dimension**—revealing how flight delays evolved across months and years under different weather conditions from 2019 to 2024. This chart enables viewers to trace seasonal variations, recurring peaks, and anomalies that static summaries cannot capture.

Purpose and Narrative Role

Whereas the bar chart provides a snapshot comparison, the line chart visualises **temporal progression**. It supports exploration of questions such as *"Are winter months consistently more delay-prone?"* or *"Did any year show atypical spikes in wind-related disruptions?"* By adding time to the visual narrative, it transforms isolated metrics into a storyline of change—connecting weather volatility with operational reliability.

Data Encoding and Colour Logic

The chart encodes **time (2019–2024)** on the x-axis and **percentage of flights delayed** on the y-axis. Three colour-coded series distinguish the weather categories:

- **Blue – Rain**
- **Red – Wind**
- **Purple – Snow**

These hues provide categorical clarity and maintain perceptual balance. Colour intensity and contrast were tuned to ensure legibility against the translucent white chart container while remaining colour-blind accessible. Smooth cubic-spline interpolation ensures that rapid value changes appear natural without implying missing intermediate data.

Interactive Behaviour

When hovering over the chart, a dynamic **tooltip** appears—displaying the **month**, **weather type**, and **exact delay percentage**. Simultaneously, the active series gains 100 % opacity while others fade to approximately 40 %, guiding attention through a clear **focus-plus-context** interaction pattern.

Hovering over a node enlarges it smoothly while other nodes dim temporarily, visually emphasising the selected point. The tooltip displays the **month, weather condition, and delay percentage**, while the annotation box beside the chart automatically updates with a short insight statement (e.g., "Rain-related delays peaked in December 2023").

Two key interactive controls enhance exploration:

1. A **dropdown menu** at the top allows users to filter by **airport** (*ATL*, *EWR*, *JFK*, or *All*). Changing this selection instantly synchronises all linked charts in real time, ensuring consistent narrative flow.
2. A **monthly selector** enables users to isolate any specific month across the 2019–2024 timeline. When a month is changed, every connected visualisation—including the grouped bar chart and heatmap—updates automatically, reinforcing the project's **interlinked dashboard design** and promoting temporal comparison within and across airports.

This interaction model not only improves usability but also supports **progressive data exploration**, encouraging users to move from overall trends to fine-grained month-by-month insights without leaving the visual context.

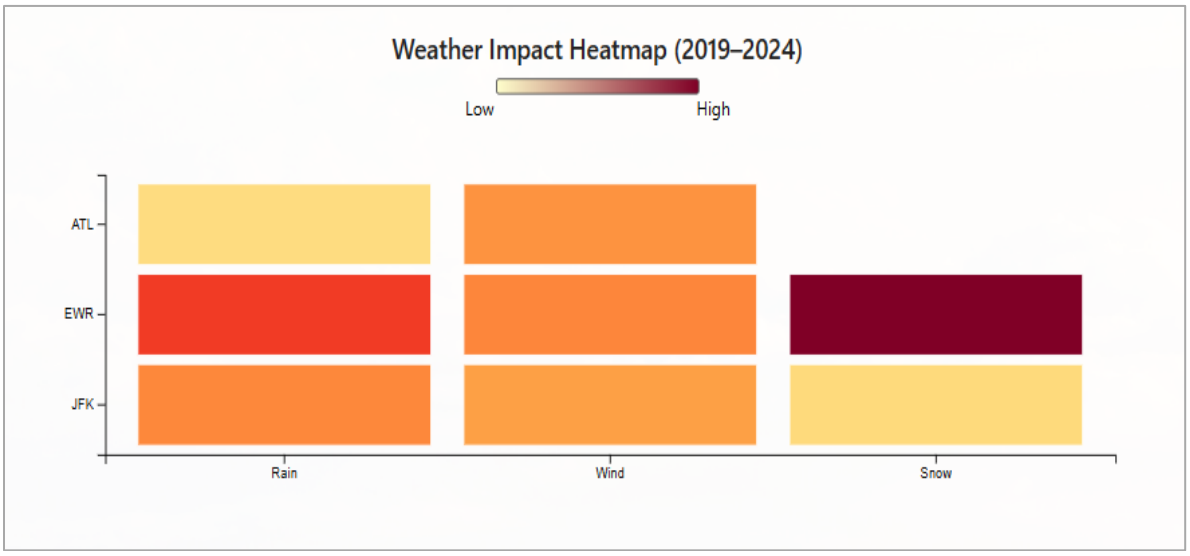
Design and Layout Considerations

The title *"Monthly Weather Trend (2019–2024)"* succinctly frames both scope and temporal range. Minimalist gridlines and light axes reduce clutter, while the white-space padding around the plot balances density with readability. Axis scales automatically adjust to preserve aspect ratios across screen sizes, ensuring responsive visual storytelling.

Summary

The line chart transforms aggregated percentages into **temporal insight**, uncovering cyclical behaviour across seasons and years. It forms the second stage of the narrative sequence: after users identify which airport–weather pairings are most critical in the bar chart, they can explore *when* those effects are most pronounced here. The coordinated transitions between these visuals deepen analytical understanding while keeping the experience visually cohesive.

3.2.3 Heatmap – Weather Impact Matrix (2019–2024)



(Figure 3.2.3 — Heatmap: Weather Impact Matrix 2019–2024.)

The **heatmap** in **Fig 3.2.3** functions as the synthesis layer of the narrative, integrating the comparative and temporal insights from the previous charts into a concise overview of **weather impact intensity** across airports. It provides a spatially structured summary of delay severity, enabling users to identify which airport–weather combinations have historically exhibited the highest or lowest disruption levels.

Purpose and Narrative Role

While the bar chart (Figure 3.2.1) highlights *which* airports are most affected and the line chart (Figure 3.2.2) reveals *when* delays occur, the heatmap answers *how severely* each weather condition influences each airport on average. Its grid-based design supports rapid cross-referencing and visual ranking, contributing to a holistic understanding of the dataset.

Data Encoding and Colour Logic

The x-axis represents **weather types** (*Rain*, *Wind*, *Snow*), and the y-axis lists the three major airports (**ATL**, **EWR**, and **JFK**). Each rectangular cell encodes **delay percentage** through a diverging colour scale—ranging from **light orange (low delay)** to **deep red (high delay)**. This palette was chosen for its perceptual linearity and intuitive "heat" metaphor, facilitating quick pattern recognition without requiring numeric precision.

A **horizontal colour legend** positioned above the chart reinforces interpretability, labelled "Low" to "High" to clarify delay magnitude. The design follows Ware's (2013) perceptual principles, ensuring the colour progression conveys intensity uniformly.

Interactive Behaviour

Each cell supports an **interactive tooltip**, which dynamically displays the airport, weather condition, and exact delay percentage when hovered. For example, hovering over the EWR–Wind cell reveals *"EWR – Wind: 26.9 % delayed compared to JFK."* This granular feedback allows users to quantify observations without disrupting the visual overview.

Hover selection triggers a **gentle highlight effect** that slightly enlarges the hovered cell's border, drawing attention without altering its colour intensity. Adjacent cells remain visible to retain full contextual awareness. This subtle motion cue preserves perceptual balance and continuity while aligning with the overall **focus-plus-context** interaction model applied consistently across all visualisations.

Annotation Box

A **contextual annotation box** adjacent to the chart summarises the key finding for viewers who prefer guided interpretation. For instance, it highlights that *for eg. EWR recorded 57.1% delays under snow* .The annotation is auto-updated during interaction, maintaining consistency with the narrative cues provided in the other charts. This feature is there with all charts.

Design and Layout Considerations

The minimalist grid design avoids unnecessary axes or gridlines, emphasising proportional cell size and colour contrast. Equal-sized rectangles maintain a balanced aspect ratio, ensuring perceptual comparability between airports and weather types. The concise title *"Weather Impact Heatmap (2019–2024)"* communicates scope immediately while aligning stylistically with the dashboard's typographic hierarchy.

Summary

The heatmap concludes the visual narrative by compressing the dataset into a compact yet information-rich matrix. It bridges the categorical clarity of the bar chart and the temporal depth of the line chart, providing users with a final comparative overview that strengthens interpretive coherence across all visuals.

4. Using the Implementation

The interactive narrative visualisation is implemented as a **self-contained HTML–CSS–JavaScript project** using **D3.js (v7)**. It runs locally on any modern browser (e.g., Chrome, Edge, and Firefox) without additional installation. The project consists of three interlinked charts—**a grouped bar chart, a monthly trend line chart, and a weather impact heatmap**—which together form a coordinated narrative illustrating the relationship between weather and flight delays in major U.S. airports.

Running the Visualisation

1. **Local Setup:**
   - Place all project files (index.html, style.css, main.js, and the dataset flights_weather_JFK_EWR_ATL_clean.csv) in a single folder.
   - Open index.html in a modern browser. **Using the local host only ** after you open it from Visual studio or any other app ull get an local host id different from for different devices do not directly open the index without the local host id it may sometime not display the charts!
   - Ensure the dataset file path is correctly linked in the JavaScript file.
2. **Dependencies:**
   - The project uses only **D3.js v7** and built-in browser capabilities.
   - No database connection is required—making it lightweight and fully portable.

User Interaction Guide

Once opened, the dashboard sequentially animates the appearance of all three charts using smooth **three-second cubic transitions**.During this brief period, users should **wait for the transitions to complete** before interacting with hover-based features, as tooltips and highlight effects become fully responsive only after the animations finish rendering.

Airport and Weather Selection

- A dropdown menu allows users to switch between **ATL**, **EWR**, **JFK**, or **All** airports.
- Weather options can be filtered through preset buttons ("Rain — Preset", "Wind — Preset", "Snow — Preset") or through individual toggles for custom combinations.
- Any change made in one control panel is immediately reflected in all charts, ensuring **real-time synchronisation**.

Interactive Behaviour

Each chart supports **hover-based exploration** and contextual annotations:

- **Grouped Bar Chart (Figure 3.2.1):** Hovering over a bar reveals the airport, weather condition, and exact percentage of delayed flights. The selected bar slightly rises and gains a border highlight, while other bars fade, creating a clear **focus-plus-context** effect.
- **Monthly Trend Line Chart (Figure 3.2.2):** Hovering over a node enlarges it smoothly while other nodes dim temporarily, visually emphasising the selected point. The tooltip displays the **month, weather condition, and delay percentage**, while the annotation box beside the chart automatically updates with a short insight statement (e.g., "Rain-related delays peaked in December 2023").Users can also use the **month selector** to isolate specific months across the 2019–2024 period, dynamically updating all linked charts.
- **Heatmap (Figure 3.2.3):** Each cell represents an airport–weather pair. Hovering over a cell gently enlarges its border (without changing its colour) and activates a tooltip showing precise delay percentages. A dedicated annotation box summarises key observations, such as higher snow-related disruptions at northern airports.

System Feedback and Interaction Timing

To preserve smooth transitions, all animations—bar rise, line draw, and heatmap fade-in—run for approximately **three seconds**. Hover interactions and tooltips activate immediately after these transitions conclude. Users are advised to wait until the animations finish before interacting, ensuring that all visual elements are fully loaded and responsive.

Summary of User Flow

1. Wait for the full chart transitions to finish loading.
2. Start with a weather preset (e.g., *Rain — Preset*) to view one condition's impact.
3. Switch airports using the dropdown for comparative insight.
4. Explore individual months through the monthly selector.
5. Hover over any bar, node, or heatmap cell to reveal exact delay values and observe contextual highlights.

These interactions encourage users to progressively explore data—moving from high-level patterns to detailed comparisons—while maintaining an intuitive, cohesive storytelling experience.

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5. Conclusion

This project successfully demonstrates how **narrative visualisation** can transform complex aviation and meteorological datasets into a coherent and engaging story. By integrating the **Bureau of Transportation Statistics (BTS)** flight performance data with **NOAA's** weather observations, the visualisation reveals the intricate and often overlooked dependency between meteorological factors and flight punctuality across major U.S. airports—JFK, EWR, and ATL. The final dashboard presents this complexity through three synchronised views—**bar, line, and heatmap charts**—allowing users to intuitively explore delay behaviour under varying conditions of rainfall, wind, and snowfall. This coordination between visuals bridges the gap between descriptive statistics and experiential understanding, turning data into an accessible narrative.

From a **design and theoretical perspective**, the visualisation adheres closely to Munzner's (2014) *What–Why–How* framework, ensuring alignment between analytical goals and perceptual tasks. The application of **consistent colour encoding**, **focus-plus-context interactivity**, and **progressive transitions** was guided by Ware's (2013) perceptual design principles and Tufte's (2001) emphasis on minimalism and visual integrity. Each chart was complemented with **annotation boxes** and **dynamic tooltips**, maintaining semantic clarity while supporting interpretive depth. The inclusion of linked filtering and fading effects across charts further reinforced cognitive continuity—an essential aspect of narrative coherence (Segel & Heer, 2010).

Technically, the project utilised **D3.js (v7)** to implement scalable SVG-based graphics with responsive design, smooth transitions, and layered interactivity. The decision to apply three-second easing animations ensured perceptual fluency and temporal balance, giving users time to cognitively register the data before exploring interactions. Developing synchronised chart updates—where a change in one view (e.g., airport selection or weather toggle) instantaneously affects all others—was among the most challenging and rewarding aspects. This design reinforced the project's narrative goal: demonstrating interdependence between environmental variability and human infrastructure systems. Through iterative debugging and visual refinement, the project evolved into a stable, performance-optimised application that runs seamlessly across devices.

Reflecting on the overall process, the most significant learning outcome was understanding how **interactivity acts as a narrative amplifier**—transforming static data interpretation into an exploratory experience. The project fostered a deeper appreciation for how theoretical concepts like cognitive load, colour perception, and temporal encoding can directly influence user engagement and message retention. In future iterations, the framework could incorporate **real-time flight API data** or **geospatial mapping overlays** for more situational awareness and live delay monitoring. Additionally, implementing accessibility features (such as high-contrast palettes or keyboard navigation) could broaden usability. Ultimately, this project achieved its objective: to communicate how weather profoundly shapes the rhythm and reliability of air travel, transforming raw data into a story that connects human experience with environmental systems through design, theory, and interaction.

6. Appendix

Figure A1. Design Sheet 1 – Brainstorming Concepts

This sheet captured the initial ideation phase, exploring multiple visualization possibilities such as geospatial maps, bubble charts, and heatmaps. While these early sketches helped identify potential data representation forms, the team later decided against geospatial mapping due to data granularity and performance limitations.

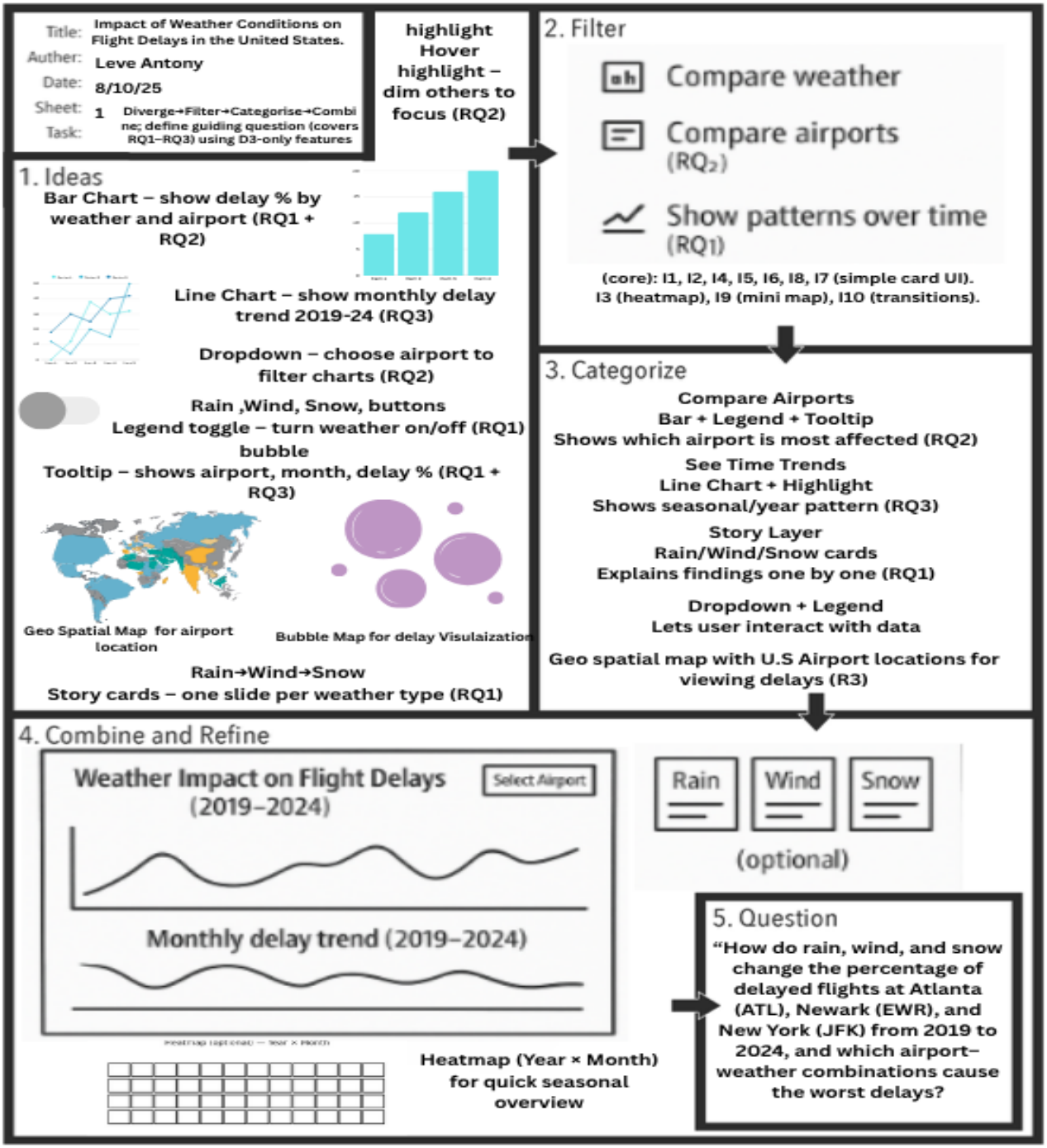


Figure A2. Design Sheet 2 – Early Layout and Heatmap Exploration

This version focused on implementing a structured dashboard layout. It introduced the idea of visual comparison across airports using grouped bars and included early concepts of color-coded heatmaps for depicting delay intensity. A consistent color palette and proportional encoding were proposed to maintain visual balance.

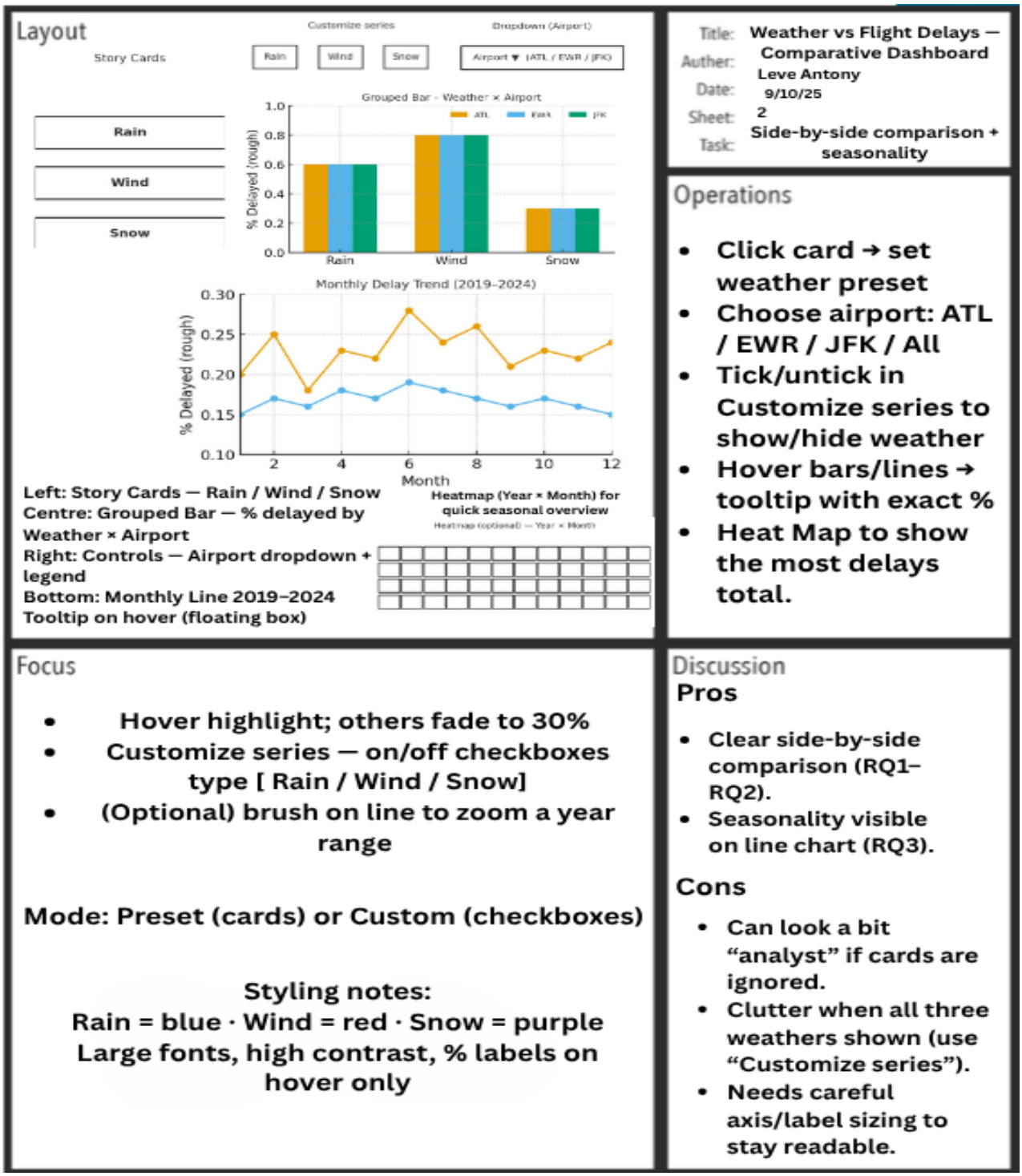


Figure A3. Design Sheet 3 – Annotation and Context Integration

This iteration added the **annotation box** concept — enabling contextual storytelling alongside charts. Each visualization included a dynamic annotation explaining trends and patterns. It also introduced hover-based tooltips and synchronized highlighting, enhancing user engagement through micro-interactions.

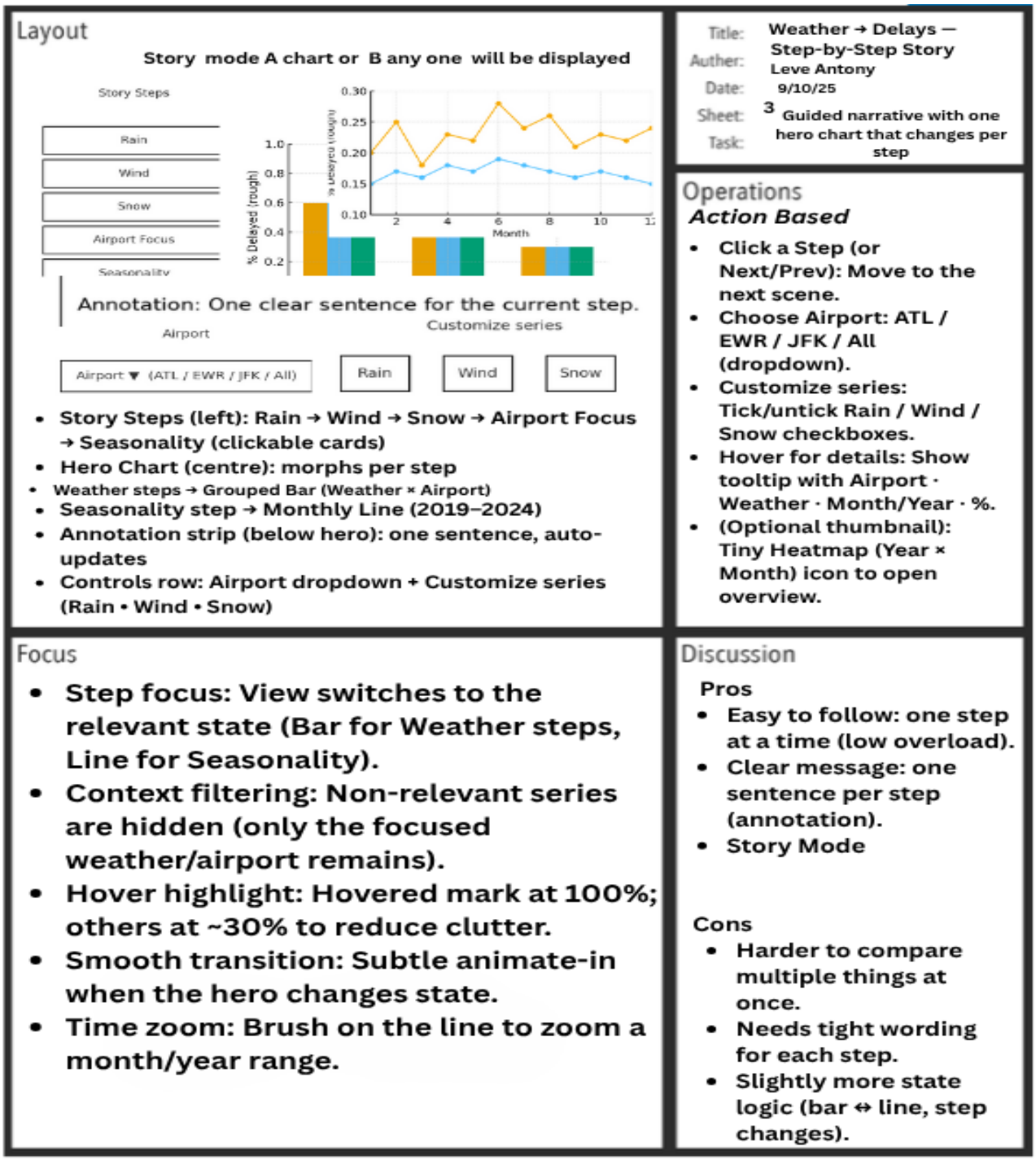


Figure A4. Design Sheet 4 – Drill-Down and Source Display

This sheet explored **drill-down interactions** and introduced **data source and description tiles** that contextualize the visualization for users. The idea was to let viewers trace delay trends from high-level summaries to granular monthly breakdowns, supporting a narrative-driven flow.

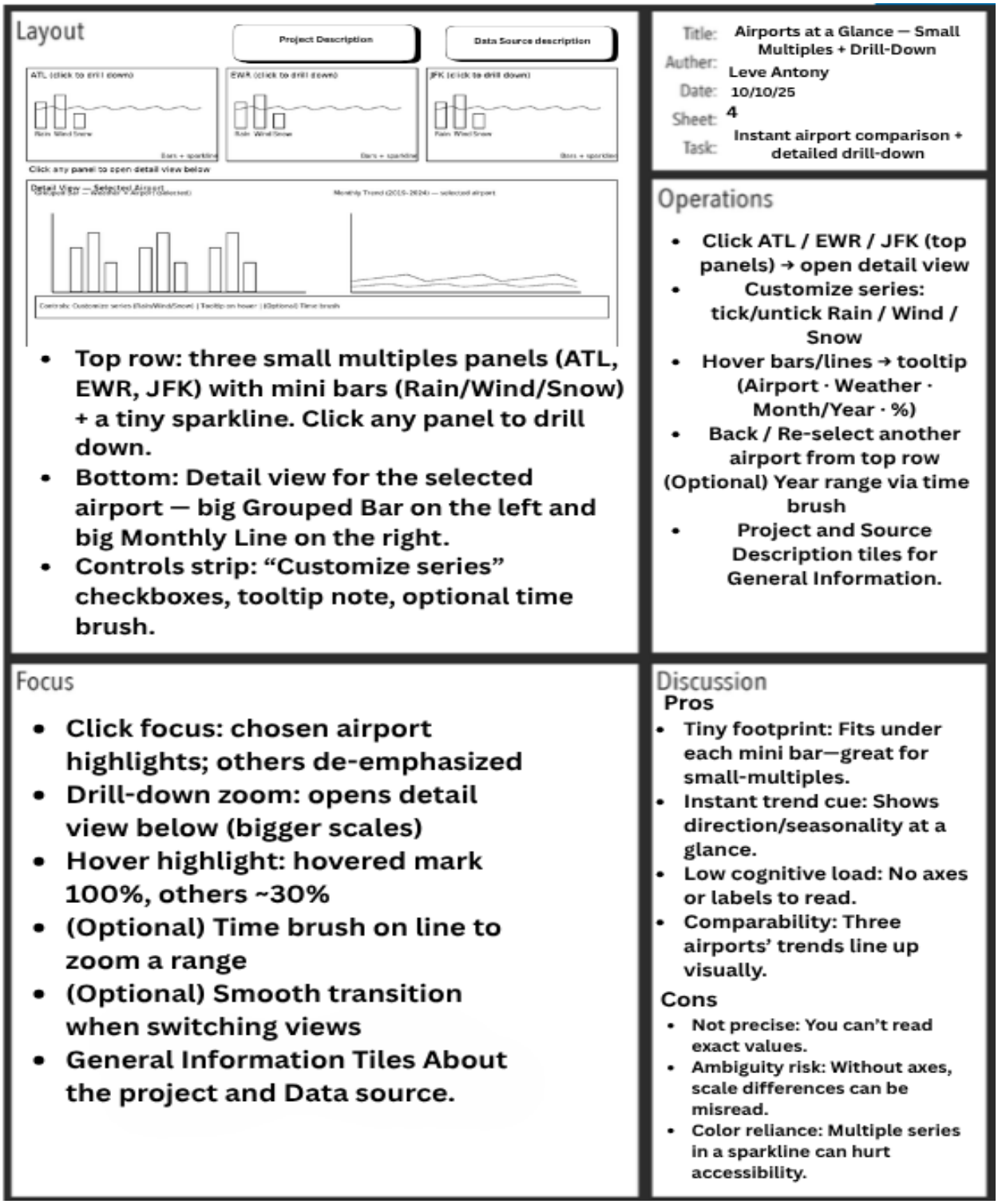
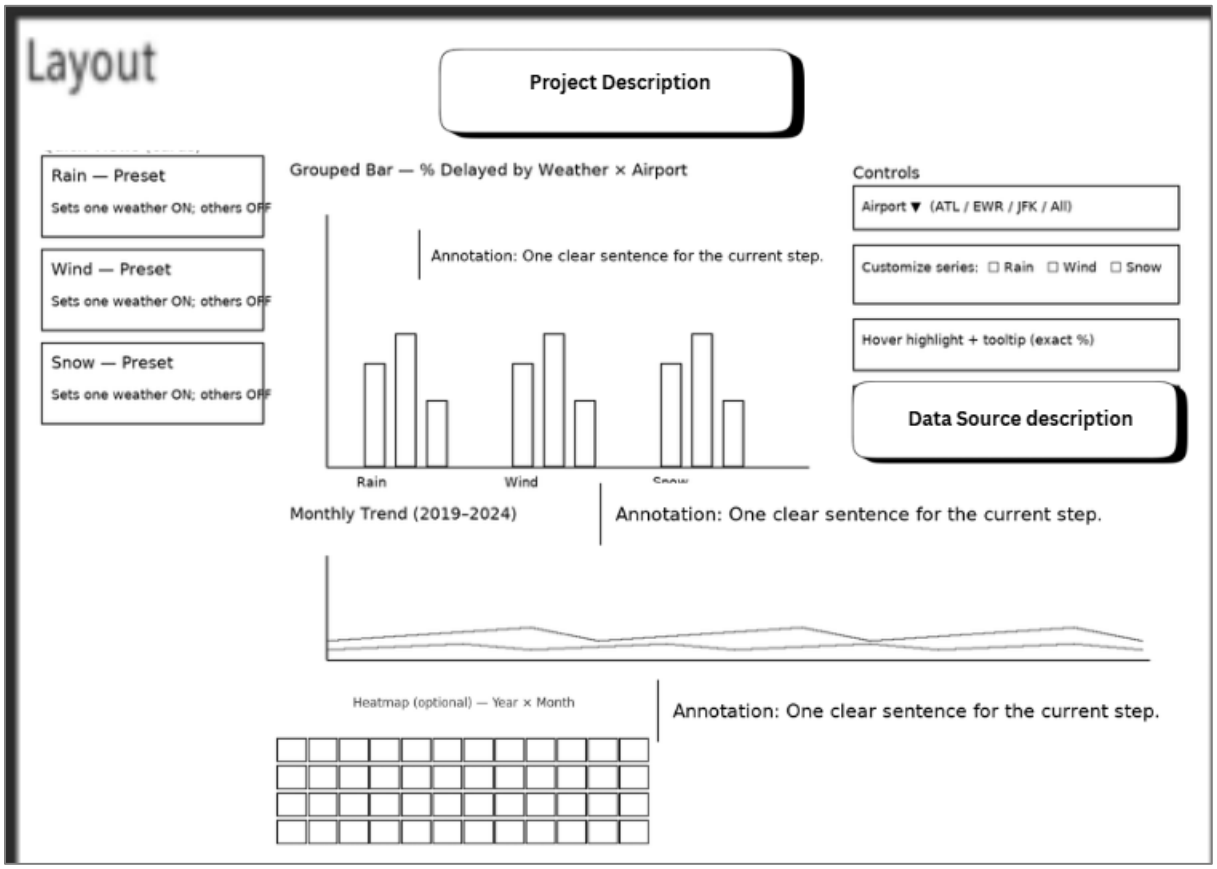


Figure A5. Design Sheet 5 – Final Integrated Design

The final design synthesized the best elements from the earlier sheets:

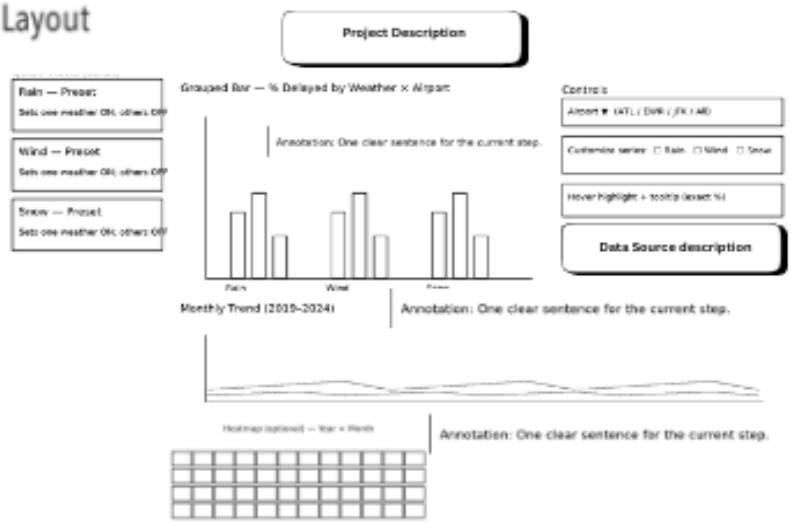
- **quick-view presets** and **annotation boxes** from Sheet 3
- **Core layout and heatmap encoding** from Sheet 2
- **Data source and description tiles** from Sheet 4

This version represented the completed dashboard layout — integrating interactivity, visual coherence, and narrative guidance to form the final implementation



Full Sheet Bellow

Layout



- Left: Quick Views (cards) — Rain / Wind / Snow (preset filters)
- Center: Grouped Bar — % delayed by Weather × Airport (RQ1 + RQ2)
- Right: Controls — Airport dropdown (ATL/EWR/JFK/All) + Customize series (on/off)
- Bottom: Monthly Line (2019–2024) (RQ3)
- Tooltip + Hover highlight across charts
- tiny Heatmap (Year × Month) if time permits
- Anotation box for better explanation where u hover with hover isnght deature legends

Title: Weather vs Flight Delays — Final Dashboard

Auther: Leve Antony

Date: 10/10/25

Sheet: 5 Based on: Sheet 2 (core) + Sheet 3 (cards)

Task: + sheet 4 project, Data Source tiles

Operations

- Estimates of Cost
- Core: 12–16 h
- Bars+Line+Tooltip: 6–8 h
- UI (cards/dropdown/checkboxes): 3–4 h
- Styling : 2 h
- Testing buffer: 1–2 h
- Optional: +2–4 h (heatmap/brush)
- Total: 12–16 h (core), up to 18–20 h with options.
- Option B — Time + $ (still tiny)
- Estimates of Cost
- Core: 12–16 h (tools/hosting: $0)
- Optional: +2–4 h
- If priced (illustrative): 12–16 h × $20–$50/h → $240–$800 (up to $1,000 with options).

Focus

Data model (what the data looks like)

- One tidy table with columns:
- airport, year, month (1–12), weather (Rain / Wind / Snow), delay_pct (percentage of flights delayed).

What the app does to the data (in the browser)

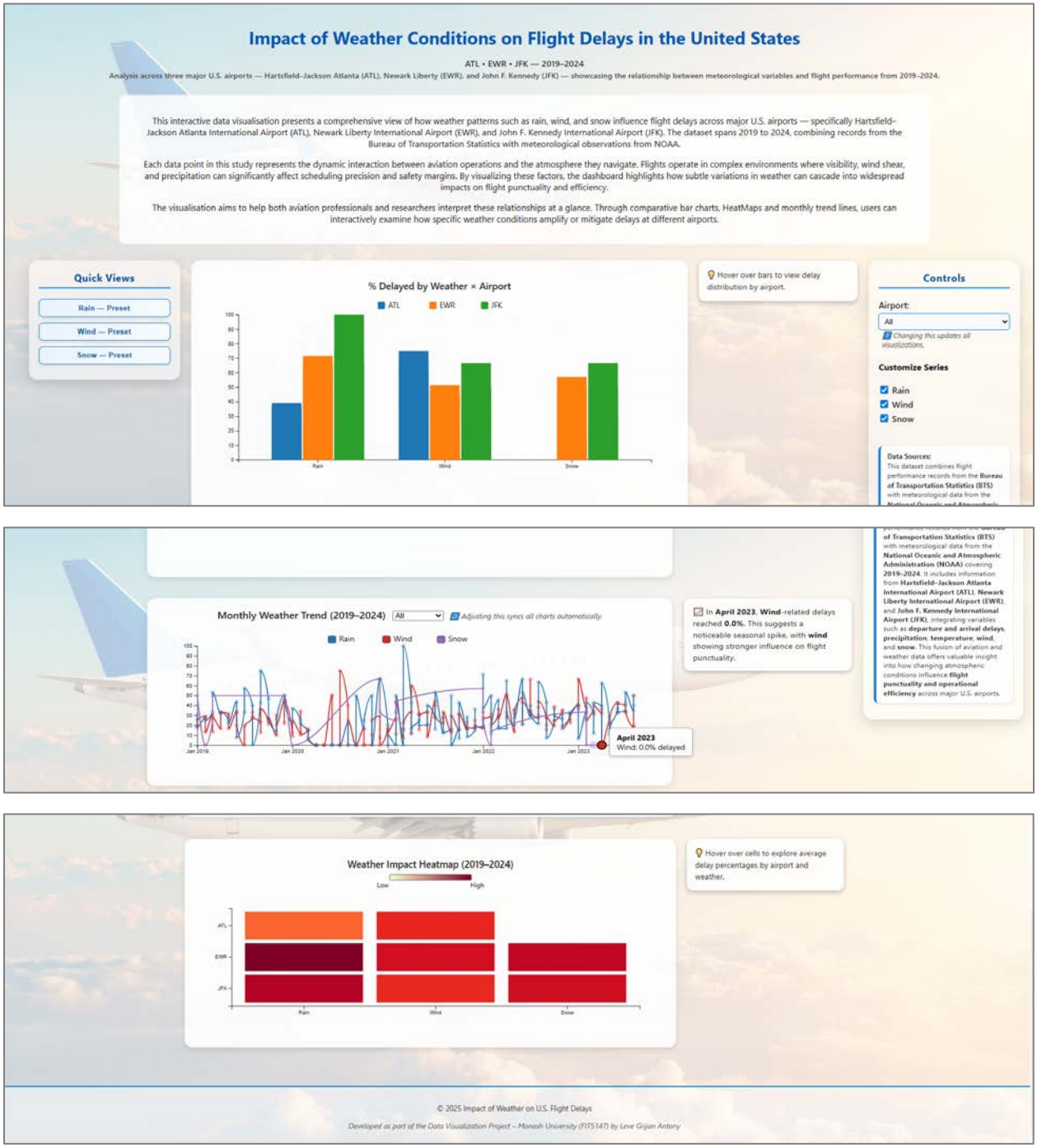
- Filter by airport (from the dropdown).
- Filter by weather (using the cards or the checkboxes).
- For the line chart, we also make a simple date from year + month so time runs left→right correctly.
- Heat Map for all the delays with highest delays
- All charts sync with each other for great visualization.

Detail

- Good for Multilevel work comparison
- Two core charts (Grouped Bar + Monthly Line) with cards, dropdown, Customize series, tooltip, hover highlight
- Guiding question and RQ1–RQ3 directly answered on a single page
- Heatmap overview with details.
- Anotation Box for detailed explanation with legend next to charts
- All charts are interlinked

Figure A6. Full Dashboard Implementation – Final Interactive View

This figure presents the **final integrated dashboard**, showing all three linked charts (bar, line, and heatmap) in a synchronized layout. Each component is dynamically connected — interactions such as hovering, airport selection, and monthly filtering trigger real-time updates across all charts. Annotation boxes on the side summarize insights, ensuring a cohesive narrative flow throughout the visualization.



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